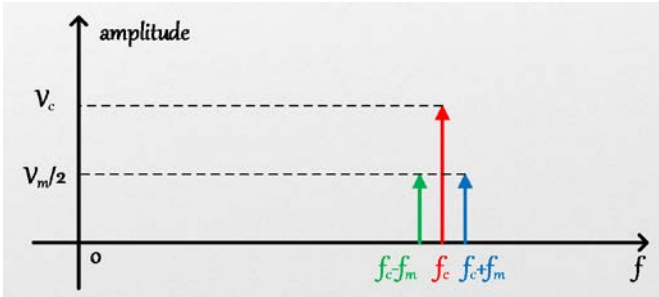
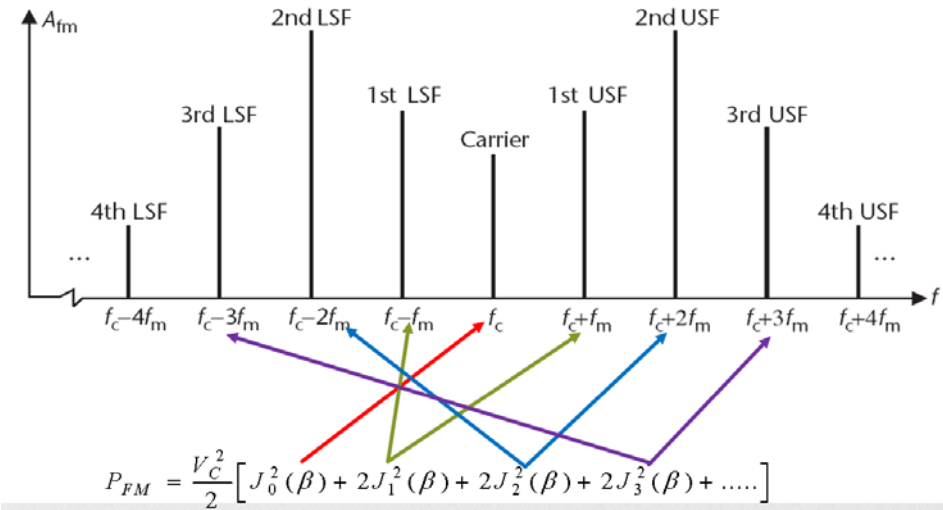


For a single-tone message signal $v_m(t) = V_m \cos(2\pi f_m t)$ and carrier signal $v_c(t) = V_c \cos(2\pi f_c t)$

	AM	FM
Expression	$[V_c + V_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$	$v_{fm}(t) = V_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)]$
Modulation Index	$m = \frac{V_m}{V_c}$	$\beta = \frac{f_d}{f_m}$
Power	$P_{t,AM} = \frac{V_c^2}{2} \left(1 + \frac{m^2}{2} \right)$	$P_{t,FM} = \frac{V_c^2}{2} = \frac{V_c^2}{2} [J_0^2(\beta) + 2J_1^2(\beta) + 2J_2^2(\beta) + 2J_3^2(\beta) + \dots]$
Spectrum	 The AM spectrum diagram shows amplitude on the vertical axis and frequency f on the horizontal axis. The origin is marked 'o'. The carrier frequency f_c is the center. Three spectral components are shown: a central carrier component at f_c with amplitude V_c (indicated by a red arrow), and two sideband components at $f_c - f_m$ (green arrow) and $f_c + f_m$ (blue arrow), both with amplitude $V_m/2$. The sidebands are labeled $f_c - f_m$, f_c, and $f_c + f_m$ on the frequency axis.	 The FM spectrum diagram shows amplitude A_{fm} on the vertical axis and frequency f on the horizontal axis. The carrier frequency f_c is the center. The spectrum consists of a central carrier component and an infinite number of sideband components. The sidebands are labeled as Lower Side Frequencies (LSF) and Upper Side Frequencies (USF). The 1st LSF is at $f_c - f_m$, the 2nd LSF is at $f_c - 2f_m$, the 3rd LSF is at $f_c - 3f_m$, and the 4th LSF is at $f_c - 4f_m$. Similarly, the 1st USF is at $f_c + f_m$, the 2nd USF is at $f_c + 2f_m$, the 3rd USF is at $f_c + 3f_m$, and the 4th USF is at $f_c + 4f_m$. The carrier component is labeled 'Carrier' at f_c. Colored arrows (red, green, blue, purple) point from the carrier and sideband components to the power equation below. The power equation is: $P_{FM} = \frac{V_c^2}{2} [J_0^2(\beta) + 2J_1^2(\beta) + 2J_2^2(\beta) + 2J_3^2(\beta) + \dots]$
Bandwidth	$B = 2f_m$	$B_c = 2(f_d + f_m)$